

TARGET PAPER SERIES — 2026

G.C.E. (A/L) COMBINED MATHEMATICS I — PURE MATHEMATICS

UNIT: INTEGRATION (CALCULUS)

STRUCTURED ESSAY QUESTIONS

TIME: 1.5 HOURS

QUESTION 01 — Integration by Parts & Reduction Formula

- (a) Using the method of integration by parts, evaluate the definite integral:

$$\int_0^{\pi/4} x \sec^2 x \, dx$$

- (b) Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$ for $n \geq 2$.

- (i) Prove that $I_n + I_{n-2} = \frac{1}{n-1}$. (ii) Hence, deduce the value of I_5 .

QUESTION 02 — Rational Functions & Partial Fractions

Consider the integrand $f(x) = \frac{2x^2 + 3x}{(x-1)(x^2+1)}$.

- (a) Express $f(x)$ in partial fractions of the form:

$$f(x) = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

- (b) Hence, evaluate the indefinite integral $\int f(x) \, dx$.

QUESTION 03 — Trigonometric Substitution & Transformations

- (a) Using the substitution $x = 2 \sin \theta$, evaluate: $\int_0^{\sqrt{2}} \frac{x^2}{\sqrt{4-x^2}} \, dx$

- (b) Show that: $\int \frac{1}{1 + \sin x + \cos x} \, dx = \ln |1 + \tan(x/2)| + C$

(Hint: Use the substitution $t = \tan(x/2)$).

QUESTION 04 — Symmetry Properties of Definite Integrals

- (a) By using property $\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx$, prove that:

$$\int_0^{\pi/2} \frac{\sin \sqrt{x}}{\sin \sqrt{x} + \cos \sqrt{x}} \, dx = \frac{\pi}{4}$$

- (b) Evaluate: $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$

QUESTION 05 — Summation Limits & Algebraic Integration

(a) Express the limit of summation as a definite integral and evaluate:

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2 + r^2}$$

(b) Using substitution $u = \sqrt{x+1}$, find: $\int \frac{\sqrt{x+1}}{x} dx$